

Tribhuvan University

Institute of Engineering

Pulchowk Campus



Final Project Report

On

LoadGate

A Concurrent L4/L7 Load Balancer Written in Go

Submitted By:

Prabesh Bastola (PUL080BCT052)

Sachin K.C. (PUL080BCT070)

Satyam Lamsal (PUL080BCT078)

Submitted To:

Department of Electronics and Computer Engineering

Subject: Network and System Programming

Instructor: Prabin Dhakal

August 2026

Declaration

We hereby declare that the work presented in this report titled *LoadGate: A Concurrent L4/L7 Load Balancer Written in Go* is our own work, carried out for the subject Network and System Programming. All sources of information and prior work used in this project have been duly acknowledged. The design, implementation, testing, and results reported here are the outcome of our own effort, and this report has not been submitted elsewhere for any other purpose.

Prabesh Bastola (PUL080BCT052)

Sachin K.C. (PUL080BCT070)

Satyam Lamsal (PUL080BCT078)

Approval

This is to certify that the project titled *LoadGate: A Concurrent L4/L7 Load Balancer Written in Go* has been examined and approved as partial fulfilment of the requirements for the subject Network and System Programming, Department of Electronics and Computer Engineering, Pulchowk Campus.

.....
Prabin Dhakal
Instructor, Network and System Programming
Department of Electronics and Computer Engineering

Copyright

The authors grant the Department of Electronics and Computer Engineering, Pulchowk Campus, permission to make this report freely available for inspection. The authors reserve other publication rights, and neither the report nor extensive extracts from it may be reproduced without the authors' written permission. The source code of LoadGate is released as open source and is available at <https://github.com/Sachinxml/loadgate>.

Abstract

LoadGate is a load balancer implemented in Go that distributes client traffic across a pool of backend servers. It operates in two modes: Layer 4 (TCP), a transparent byte-stream proxy compatible with any TCP-based protocol, and Layer 7 (HTTP/HTTPS), a reverse proxy capable of path- and header-aware routing. The system provides a unified load-balancing interface with four algorithms — Round Robin, Smooth Weighted Round Robin, Least Connections, and Consistent Hashing with virtual nodes — together with a two-plane health-checking system, automatic failover with request retry, TLS termination with hot certificate rotation, zero-downtime configuration reload, token-bucket rate limiting, and Prometheus/Grafana observability.

The project was evaluated against five measurable goals: no goroutine leaks under fault conditions, no dropped in-flight requests during reload and shutdown, sub-second failover, live observability, and a reproducible benchmark report. The measured results show that the proxy adds under half a millisecond of latency per request, scales linearly and cheaply to ten thousand concurrent connections (under 200 MB of memory), moves only $1/N$ of keys when a backend is removed under consistent hashing, and serves every client request with no error burst across repeated backend failures. Chaos testing revealed three real correctness bugs in the failover path, which were diagnosed and fixed; the debugging process itself is documented as a contribution of this work.

Acknowledgements

We would like to express our sincere gratitude to our instructor, Prabin Dhakal, for his guidance throughout this project, and to the Department of Electronics and Computer Engineering, Pulchowk Campus, for providing the opportunity to undertake it. We are also grateful to the open-source community whose tools — the Go programming language, Prometheus, Grafana, and Docker — made this work possible.

Contents

Declaration	i
Approval	ii
Copyright	iii
Abstract	iv
Acknowledgements	v
Abbreviations	ix
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Objectives	2
1.3.1 General Objective	2
1.3.2 Specific Objectives	2
1.4 Rationale	2
1.5 Scope	2
1.6 Limitations	3
1.7 Outline of the Report	3
2 Literature Review	4
2.1 Review of Related Theory	4
2.2 Review of Related Works	5
2.3 Gap Analysis	5
3 Methodology	6
3.1 Project Approach	6
3.2 System Architecture	6
3.3 System Design and Modules	6
3.4 Technology Stack	7
3.5 Development Methodology	7
3.6 Data Handling	7
3.7 Testing and Validation	8
3.8 Deployment Setup	8
4 Implementation Details	9
4.1 Requirements Specification	9
4.2 System Overview	9

4.3	Development Environment	9
4.4	System Implementation	10
4.4.1	The request pipeline	10
4.4.2	L4 — the TCP relay	10
4.4.3	L7 — correct HTTP forwarding	10
4.4.4	The four algorithms	10
4.4.5	The health state machine	10
4.4.6	Failover with retry	11
4.4.7	Configuration reload and TLS	11
4.4.8	Rate limiting and metrics	11
4.5	Interface and Visualization	11
4.6	Deployment and Execution	12
4.7	Challenges and Solutions	12
5	Results and Discussion	13
5.1	Results	13
5.2	Discussion	14
5.3	Comparative Analysis	15
6	Conclusions and Recommendations	16
6.1	Conclusions	16
6.2	Recommendations and Future Enhancements	16
	References	17
A	Configuration Reference	18
B	Command-Line Reference	19
C	Reproducing the Results	20

List of Figures

3.1	High-level architecture of LoadGate. Solid arrows are the client-request path; dashed arrows are configuration updates and monitoring.	6
4.1	Backend health state machine. Eviction is immediate; recovery requires a cooldown and several consecutive good probes.	11

List of Tables

2.1	Comparison of Layer 4 and Layer 7 load balancing.	4
2.2	LoadGate in relation to existing load balancers.	5
3.1	Technology stack.	7
4.1	Development and measurement environment.	9
4.2	Default health-check parameters (all configurable).	11
4.3	Key Prometheus metrics.	12
5.1	E1: per-request overhead of the proxy.	13
5.2	E2: goroutine-per-connection scaling.	13
5.3	E3: algorithm behaviour with two 5 ms and one 150 ms backend.	13
5.4	E4: fraction of keys moved when one of ten backends is removed.	14
5.5	E5: latency with and without passive eviction, one sick backend.	14
5.6	Results against the project goals.	14

Abbreviations

L4	Layer 4 (transport layer, TCP)
L7	Layer 7 (application layer, HTTP)
TCP	Transmission Control Protocol
HTTP	Hypertext Transfer Protocol
HTTPS	HTTP Secure
TLS	Transport Layer Security
RR	Round Robin
WRR	Weighted Round Robin
XFF	X-Forwarded-For (HTTP header)
SNI	Server Name Indication
RPS	Requests Per Second
RSS	Resident Set Size (process memory)
GC	Garbage Collection
LRU	Least Recently Used
FNV	Fowler–Noll–Vo (hash function)
RTT	Round-Trip Time
CI	Continuous Integration
YAML	YAML Ain't Markup Language
p50/p99	50th / 99th percentile latency

Chapter 1

Introduction

1.1 Background

Modern network services rarely run on a single server. To handle load and to survive individual failures, a service is deployed as a pool of identical backend servers, and a *load balancer* sits in front of that pool. The load balancer presents a single stable address to clients and spreads incoming traffic across the backends. This gives a service three things a single server cannot provide: aggregate capacity, high availability (traffic moves away from a failed backend), and a stable entry point that hides the changing set of backends behind it.

Load balancers can operate at different layers of the network stack. A *Layer 4* balancer works at the transport layer and forwards raw TCP connections without inspecting the data inside them; it is fast and protocol-agnostic. A *Layer 7* balancer works at the application layer, understands HTTP, and can therefore make richer routing decisions based on the request path, headers, or cookies, at the cost of parsing every request. LoadGate implements both, so that the trade-off between the two can be studied directly.

Building such a system exercises the core of network and systems programming: socket programming, safe concurrency across thousands of simultaneous connections, fault detection and recovery, and operational concerns such as configuration reload, graceful shutdown, and monitoring.

1.2 Problem Statement

Without a load balancer, client requests cannot be distributed across multiple backends, and the failure of a single backend can reduce the availability of the entire service. A correct load balancer must therefore solve several hard problems at once. It must handle many concurrent connections without leaking resources; it must detect a failed backend quickly and stop sending it traffic, yet readmit it carefully once it recovers; it must survive configuration changes and its own shutdown without dropping in-flight requests; and it must do all of this with low, predictable overhead. The problem this project addresses is to design and implement such a load balancer, and to *prove*, with measurements rather than assertions, that it meets these requirements.

1.3 Objectives

1.3.1 General Objective

To design, implement, and evaluate a concurrent, production-style load balancer in Go that distributes client traffic across a pool of backend servers while maintaining high availability, efficient resource use, and fault tolerance.

1.3.2 Specific Objectives

The project is evaluated against five specific, measurable objectives:

G1 — Correct concurrency (no leaks)

The goroutine count returns to its baseline after a sustained chaos test, verified by sampling and by pprof goroutine profiles.

G2 — Zero dropped in-flight requests

Shutdown (`SIGTERM`) and reload (`SIGHUP`) under sustained load complete all already-accepted requests.

G3 — Fast failover

When a backend is killed under load, the client-visible disruption window is under one second.

G4 — Observability

Prometheus metrics and a Grafana dashboard make a backend's failure and recovery visible within one probe interval.

G5 — Reproducible performance characterization

A benchmark report with throughput, latency percentiles, and scaling data, all reproducible via `make bench`.

1.4 Rationale

The project follows three design principles. First, **quality over quantity**: a smaller set of features implemented correctly and tested thoroughly is worth more than many features that merely appear to work. Second, **everything should be measurable**: every important design decision is backed by a benchmark, a profile, or a test rather than an assumption. Third, **incremental development**: the system was built in phases, each ending in a working, demonstrable program. These principles make the final result trustworthy — each claim in this report can be re-checked by running the corresponding command.

1.5 Scope

The project delivers a single load-balancer process supporting both L4 and L7 modes, all four balancing algorithms, the two-plane health system, TLS termination with reload-based certificate rotation, hot configuration reload, rate limiting, Prometheus metrics with a Grafana dashboard, a chaos-testing harness, and a reproducible benchmark suite. Backends are declared in a YAML configuration file. A Docker Compose stack runs the balancer together with demo backends and the monitoring stack.

1.6 Limitations

The following are intentionally out of scope: HTTP/2 and gRPC to backends; service discovery via external systems such as Consul or etcd (backends are declared in the config file); running multiple cooperating load-balancer instances for high availability; and rate limiting shared across machines. In addition, all benchmarks are single-machine over loopback, so they measure the balancer’s own overhead rather than real-world network behaviour, and the active health probe checks TCP connectivity only.

1.7 Outline of the Report

Chapter 2 reviews the relevant theory and existing systems. Chapter 3 describes the methodology — the project approach, system architecture, module design, technology stack, and testing strategy. Chapter 4 covers implementation in detail, including the requirements, the core algorithms and mechanisms, and the challenges faced. Chapter 5 presents the results and discusses them, including comparative analysis. Chapter 6 concludes and lists future enhancements.

Chapter 2

Literature Review

2.1 Review of Related Theory

Layer 4 versus Layer 7 load balancing. The two operating modes work with different information and therefore offer different capabilities, summarised in Table 2.1. L4 forwards TCP connections with minimal processing; L7 parses HTTP and can route on request content.

Table 2.1: Comparison of Layer 4 and Layer 7 load balancing.

Aspect	L4 (TCP)	L7 (HTTP)
Visible information	Source/destination IP and ports	Method, URL, headers, cookies
Routing unit	One backend per TCP connection	One backend per HTTP request
Protocols	Any TCP-based protocol	HTTP and HTTPS only
Balancing key	Usually client IP	Client IP, path, header, or cookie
Overhead	Minimal (direct forwarding)	Higher (requests are parsed)
Failure detection	Connection failures only	Connection failures and HTTP status
TLS	Encrypted traffic forwarded as-is	Can be terminated at the balancer

Consistent hashing. A naive hash(*key*) mod N mapping reassigns nearly every key when N changes. Consistent hashing, introduced by Karger et al. [2], places both backends and keys on a ring so that adding or removing one backend reassigns only about $1/N$ of the keys. Virtual nodes (multiple ring positions per backend) smooth out the distribution. This makes it the standard choice for cache affinity and sticky sessions.

Smooth weighted round robin. A naive weighted scheme sends a burst of consecutive requests to a heavier backend. The smooth weighted round-robin algorithm, popularised by nginx [3], interleaves picks by tracking a running “current weight” per backend, producing a smoother distribution.

Health checking. Production systems combine *active* probing with *passive* observation of real traffic; the asymmetric “evict quickly, readmit slowly” policy mirrors outlier-detection features in Envoy and HAProxy [4].

Concurrency and the Go netpoller. Go schedules many goroutines onto a small number

of OS threads and parks a goroutine that is waiting on network I/O using an event notification mechanism (`epoll` on Linux) [5]. A goroutine starts with a small stack (a few kilobytes), so a goroutine-per-connection design remains practical at thousands of connections.

Coordinated omission. Tene [6] shows that a load generator which waits for each reply before sending the next one hides latency during stalls. Correct latency measurement requires an open-loop, constant-rate generator — a principle this project’s benchmark methodology follows.

2.2 Review of Related Works

Several mature load balancers informed this design:

- **NGINX** — a widely deployed reverse proxy and L7 load balancer; the source of the smooth weighted round-robin algorithm used here.
- **HAProxy** — a high-performance L4/L7 balancer known for its health-checking and observability; its `rise/fall` probe semantics are adopted directly.
- **Envoy** — a modern service-mesh proxy whose outlier-detection model (passive ejection plus active checks) matches the two-plane health design.
- **Traefik** — a Go-based reverse proxy and load balancer, demonstrating that the goroutine-per-connection model scales in practice.

Table 2.2 places LoadGate alongside these systems.

Table 2.2: LoadGate in relation to existing load balancers.

Feature	NGINX	HAProxy	Envoy	LoadGate
L4 and L7 modes	partial	yes	yes	yes
Consistent hashing	yes	yes	yes	yes
Active + passive health	yes	yes	yes	yes
Hot reload	yes	yes	yes	yes
Written in Go	no	no	no	yes
Scope	large	large	large	focused / educational

2.3 Gap Analysis

The existing systems are large, mature, and written mostly in C or C++. They are excellent tools but poor *teaching artefacts*: their scale hides the core mechanisms. LoadGate fills a narrower gap — a compact, readable, fully-tested load balancer written against the Go standard library, where each mechanism (the lock-free data path, the health state machine, the consistent-hash ring, the failover retry) is small enough to be understood, yet complete enough to be measured against the same properties production systems care about. The contribution is not a new algorithm but a correct, measured, and documented synthesis.

Chapter 3

Methodology

3.1 Project Approach

The system was built *incrementally* across seven phases (P0–P6), each ending in a working, demonstrable program with its own exit criterion. P0 established the skeleton (config parsing, CI); P1 the L4 TCP proxy; P2 the health loop; P3 the L7 proxy and TLS; P4 the remaining algorithms; P5 operations (reload, rate limiting, metrics, dashboard); and P6 the chaos harness, benchmarks, and this report. Correctness goals (G1–G3) were treated as non-negotiable: under schedule pressure, features would be cut before correctness tests.

3.2 System Architecture

The central architectural decision is a strict separation between a *data plane* and a *control plane*. The data plane is the hot path — accept a connection, pick a backend, move bytes — and must never block on administrative work. The control plane is the cold path — load and validate configuration, run health probes, react to signals — and may use locks freely. The two planes meet at exactly one point: an immutable configuration snapshot published behind a single atomic pointer. Figure 3.1 shows the overall structure.

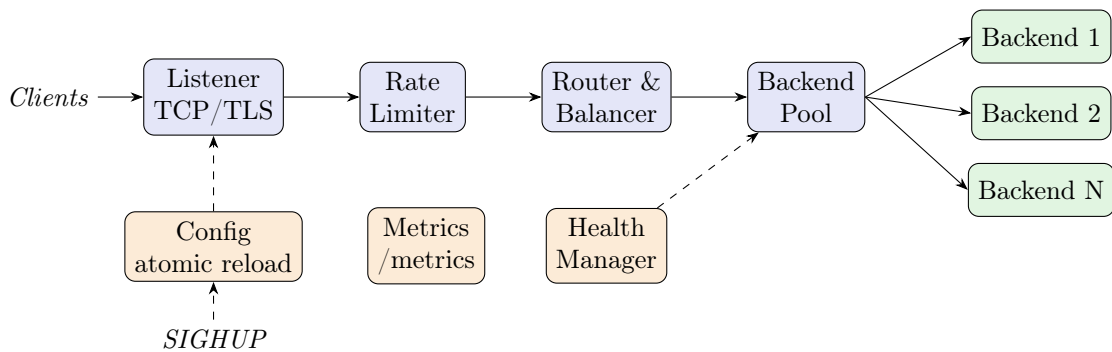


Figure 3.1: High-level architecture of LoadGate. Solid arrows are the client-request path; dashed arrows are configuration updates and monitoring.

3.3 System Design and Modules

The balancer is divided into independent modules, each with one responsibility:

- **Listener** — accepts incoming TCP or TLS connections.

- **Rate Limiter** — enforces global and per-client request limits.
- **Router and Balancer** — selects the backend pool (by path in L7) and picks a healthy backend using the configured algorithm.
- **Backend Pool / Registry** — holds each backend’s health state and live connection count, outside the config snapshot so it survives reloads.
- **Health Manager** — runs one prober per backend, evicting and readmitting backends through the state machine.
- **Configuration Manager** — loads, validates, and atomically swaps the config.
- **Metrics System** — exports Prometheus metrics on a separate port.

3.4 Technology Stack

The implementation targets Go 1.26 and deliberately limits itself to four runtime dependencies, relying on the standard library for everything else. The stack is summarised in Table 3.1.

Table 3.1: Technology stack.

Concern	Choice
Language	Go 1.26
Config parsing	<code>gopkg.in/yaml.v3</code>
Rate limiting	<code>golang.org/x/time/rate</code>
Metrics	<code>prometheus/client_golang</code>
Optional file watch	<code>fsnotify</code>
Leak detection (test)	<code>go.uber.org/goleak</code>
Containerisation	Docker, Docker Compose
Monitoring	Prometheus, Grafana
Profiling	<code>net/http/pprof</code>

3.5 Development Methodology

Development followed an incremental, test-driven process. Each package carries unit tests run under the race detector, and continuous integration runs formatting checks, `go vet`, and the full test suite on every change. Work was tracked in phases, each merged only once its exit criterion was met.

3.6 Data Handling

There are three kinds of data in the system. *Configuration* is read from a YAML file, validated, and held as an immutable snapshot behind an atomic pointer; a reload builds a new snapshot and swaps it. *Request and response bodies* are streamed, never fully buffered, so that large payloads do not inflate memory. *Operational data* (per-backend health and connection counts, metric counters) lives in small concurrent structures — atomic counters on the hot path, a mutex only where state genuinely changes.

3.7 Testing and Validation

Validation uses five complementary techniques: unit tests with the race detector; goroutine-leak detection with `goleak`; differential (equivalence) testing of the L7 proxy against the standard library's `httputil.ReverseProxy`; a chaos harness that kills backends under load; and a reproducible benchmark suite. These map directly onto the goals: the chaos harness proves G1 and G3, the reload and drain tests prove G2, the dashboard proves G4, and the benchmark suite proves G5.

3.8 Deployment Setup

Two deployment paths are used. For demonstration and chaos testing, a Docker Compose stack runs the balancer, three demo backends, Prometheus, and a preloaded Grafana dashboard; the balancer is built as a static binary into a `scratch` image via a multi-stage Dockerfile. For benchmarking, the components run as bare host processes over loopback, so that measurements reflect the balancer rather than Docker's virtual network.

Chapter 4

Implementation Details

4.1 Requirements Specification

Functional requirements. The system shall: forward TCP connections (L4) and HTTP(S) requests (L7); select a backend using any of four configurable algorithms; route L7 requests to a pool by longest matching path prefix; detect a failed backend and remove it from rotation; readmit a recovered backend after successful probes; retry a failed idempotent request on another backend; terminate TLS and rotate certificates on reload; reload configuration on `SIGHUP` without dropping connections; enforce global and per-client rate limits; and export Prometheus metrics.

Non-functional requirements. The system shall add sub-millisecond latency on the hot path; scale to at least ten thousand concurrent connections without leaking resources; never drop an accepted request during reload or shutdown; and keep metric cardinality bounded.

Constraints. At most four runtime dependencies; Go 1.26; Linux as the primary platform.

4.2 System Overview

The methodology of Chapter 3 maps onto concrete Go packages: a `config` package (parse, validate, atomic `Store`); a `balancer` package (the four algorithms behind a common `Pick` interface); a `health` package (the registry, state machine, and prober manager); `proxy/14` and `proxy/17` packages (the two data paths); a `ratelimit` package; a `reload` package; and a `metrics` package. A request reads the config snapshot once and flows through these packages without acquiring a lock on the hot path.

4.3 Development Environment

The system was developed and measured on the environment in Table 4.1.

Table 4.1: Development and measurement environment.

CPU	13th Gen Intel Core i5-13420H (12 cores)
Memory	15 GiB
OS	Linux (Pop!_OS, kernel 7.0.11)
Go toolchain	go1.26.5

4.4 System Implementation

4.4.1 The request pipeline

For an L7 request the pipeline is: (1) rate-limit check; (2) load the config snapshot with one atomic read; (3) route to a pool by longest matching path prefix; (4) filter the pool to healthy backends; (5) pick one backend by the configured algorithm; (6) forward, retrying on another healthy backend if the connection fails; (7) stream the response back. In L4 mode the pipeline is similar but the “request” is the whole TCP stream, selected once and relayed byte-for-byte.

4.4.2 L4 — the TCP relay

An accepted connection uses three goroutines: one owning the connection plus one for each direction of an `io.Copy`. The tricky part is closing correctly: when one direction ends, the code half-closes that side (`CloseWrite`, sending an EOF) so the other direction can finish, applies a deadline so a stalled peer cannot hold the connection forever, and only then closes both sides and decrements the connection count. Getting this wrong is the classic source of goroutine leaks, which is why it is exercised directly by the chaos test.

4.4.3 L7 — correct HTTP forwarding

Forwarding an HTTP request correctly means obeying four rules: clear the incoming `RequestURI` before reusing the request outbound; strip hop-by-hop headers (`Connection`, `Keep-Alive`, `Transfer-Encoding`, and any names listed inside `Connection`); append the client IP to `X-Forwarded-For`; and stream the body. Backend connections are pooled through a shared `http.Transport`. A key setting is `ResponseHeaderTimeout`, fixed at 300 ms: it bounds how long a pooled connection to a silently-dead backend waits before failing. Rather than use the standard library’s `httputil.ReverseProxy`, `LoadGate` implements the proxy directly for control over these rules, and validates the result against the library proxy (Section 4.7).

4.4.4 The four algorithms

Round Robin uses a single atomic counter modulo the pool size — cheapest, even by count, blind to load. **Smooth Weighted Round Robin** tracks a running current weight per backend under a small lock, interleaving picks so a 3:1 weighting yields A B A C A B A C. **Least Connections** picks the backend with the fewest in-flight requests (each backend carries an atomic counter incremented on dispatch and decremented on completion), breaking ties with round robin; it naturally steers away from a slow backend whose counter stays high. **Consistent Hashing** builds a ring with 150 virtual nodes per backend using an FNV-1a hash, published immutably behind an atomic pointer and rebuilt only when the pool’s membership fingerprint changes; a key maps to the first backend clockwise from its hash, so removing a backend disturbs only its own keys.

4.4.5 The health state machine

Each backend moves through four phases — Healthy, Evicted, Cooldown, Probation — following the policy “fail fast, recover carefully” (Figure 4.1). The passive path is the only mechanism that evicts (after `fall` consecutive real-traffic failures); the active prober is the only mechanism that readmits (after a cooldown and `rise` consecutive successful probes). The registry keeps this state outside the config snapshot, and a reload reconciles rather than rebuilds it. Table 4.2 lists the default parameters.

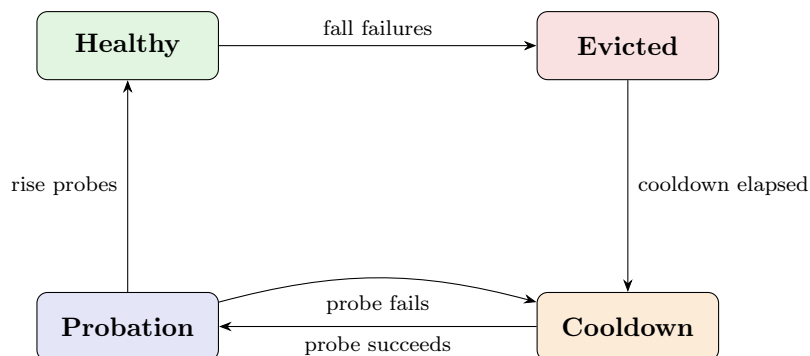


Figure 4.1: Backend health state machine. Eviction is immediate; recovery requires a cooldown and several consecutive good probes.

Table 4.2: Default health-check parameters (all configurable).

Parameter	Default	Role
fall (failure threshold)	3	passive: failures before eviction
cooldown	10s	wait after eviction before probing
rise (success threshold)	2	active: good probes before readmit
probe interval	2s	active: time between probes
probe timeout	500ms	active: max wait for a probe

4.4.6 Failover with retry

When a connection-level failure occurs and the client has not itself disconnected, the request is retried on a different healthy backend — for idempotent methods only, across up to three distinct backends, and only before any response byte has been written. This makes a dying backend invisible to clients while the health system catches up.

4.4.7 Configuration reload and TLS

Reload follows a strict validate-before-swap discipline: on `SIGHUP` the new file is parsed and validated completely, and only then is the config pointer swapped and the registry reconciled; an invalid file is rejected and the running config is kept. TLS certificates are resolved per handshake from the current snapshot, so certificate rotation is just a reload.

4.4.8 Rate limiting and metrics

Rate limiting uses token buckets at two levels (global and per-client-IP), with per-client buckets in an LRU-bounded map so a flood of unique IPs cannot exhaust memory; over-limit traffic is rejected immediately rather than queued. Metrics are exported on a separate port with bounded cardinality — labelled by status class and backend, never by exact code, client IP, or path. Table 4.3 lists the key series.

4.5 Interface and Visualization

LoadGate is operated through a command-line interface (`run` and `check` sub-commands) and a YAML configuration file (Appendix A and B). Live state is visualised through a Grafana dashboard, provisioned automatically in the Docker stack, showing request rate and latency, per-backend health and active connections, goroutine and memory usage, and rate-limit and reload events. During a demonstration the dashboard shows a backend being killed, evicted,

Table 4.3: Key Prometheus metrics.

Metric	Type	Purpose
request count	counter	requests by backend and status class
request duration	histogram	latency percentiles by backend
active connections	gauge	live connections per backend
backend up	gauge	health of each backend
health transitions	counter	evictions and readmissions
rate-limit rejections	counter	requests refused by the limiter
reload outcomes	counter	successful and failed reloads

and readmitted — satisfying goal G4. Profiling data is available on demand through pprof endpoints on a private debug port.

4.6 Deployment and Execution

The system is built with `make build` and run as `loadgate run -c config.yaml`. The full demonstration stack is started with `docker compose up -build`, exposing the proxy on port 8080, metrics on 9095, and Grafana on 3000. Benchmarks run as bare processes via `make bench`; the chaos harness runs via `bash test/chaos/run.sh`. All commands are listed in Appendix C.

4.7 Challenges and Solutions

The most instructive part of the project was a set of correctness bugs in the failover path that only surfaced under chaos testing, where killing backends under load initially produced multi-second error windows and a goroutine count climbing from 14 into the hundreds. Three distinct bugs were hiding behind that one symptom.

Challenge 1: a client hang-up counted as a backend failure. The proxy derived each request’s context from the incoming request. When a client gave up and closed its connection, that context was cancelled, the outbound call returned an error, and the code recorded a backend failure. Under load this evicted *healthy* backends — sometimes all of them at once, producing 100%-failure windows. *Solution:* check whether the client’s own context was cancelled and, if so, do not blame the backend.

Challenge 2: no retry. Any connection failure returned 502 to the client immediately, so every request that landed on a just-killed backend was a visible error. *Solution:* retry idempotent requests on another healthy backend before returning an error.

Challenge 3: dead connections were never closed. A dead backend’s pooled connections lingered until a 90-second idle timeout, each with a live goroutine, so the count climbed with every kill. *Solution:* close idle connections the moment a backend is evicted, and tighten the pool.

Two further lessons: the `goleak` check must run through `TestMain`, not a deferred call inside a test, or it races the test’s own cleanup and reports false leaks; and the L7 latency tax is sub-millisecond, so the benchmark generator had to record microseconds, not milliseconds, to make it visible. Chaos testing did not just verify the system — it *found real bugs*, which is exactly its purpose.

Chapter 5

Results and Discussion

5.1 Results

All benchmarks ran as bare processes over loopback, driven by an open-loop constant-rate generator, on the environment of Table 4.1. Each experiment is reproducible via `make bench`.

E1 — Proxy overhead. Latency direct versus through the proxy, at 1000 rps against a 5 ms backend (Table 5.1). The proxy adds under half a millisecond; L7 costs about twice L4.

Table 5.1: E1: per-request overhead of the proxy.

target	p50	p99	overhead (p50)
direct	5.59 ms	6.46 ms	—
through L4	5.78 ms	6.60 ms	+0.19 ms
through L7	6.01 ms	7.01 ms	+0.42 ms

E2 — Connection scaling. Goroutines and memory versus live connections (Table 5.2). The goroutine count is exactly $12 + 3 \times \text{connections}$ — perfectly linear — and 10,000 connections cost under 200 MB with zero failures.

Table 5.2: E2: goroutine-per-connection scaling.

connections	goroutines	memory
100	312	16.9 MB
1000	3012	32.9 MB
5000	15012	107.5 MB
10000	30012	189.5 MB

E3 — Least-connections versus round-robin with one slow backend (Table 5.3). Least-connections cuts mean latency by a factor of five by steering away from the slow backend.

Table 5.3: E3: algorithm behaviour with two 5 ms and one 150 ms backend.

algorithm	p50	p90	mean
round_robin	6.5 ms	151 ms	54.7 ms
least_connections	6.2 ms	6.9 ms	10.1 ms

E4 — Consistent-hash remap (Table 5.4). Removing one of ten backends moves exactly $1/N$ of keys under consistent hashing, versus 90% for naive modulo.

Table 5.4: E4: fraction of keys moved when one of ten backends is removed.

method	keys moved (avg)	per-drop range
consistent_hash	10.0%	4.3%–16.0%
modulo_n	90.1%	—

E5 — Value of the health loop (Table 5.5). With a sick backend, passive eviction keeps the mean near 8 ms; disabling it lets a third of requests suffer the sick backend, pushing the mean to 106 ms.

Table 5.5: E5: latency with and without passive eviction, one sick backend.

health	p50	p90	mean
passive+active	6.1 ms	6.7 ms	8.3 ms
active-only	6.5 ms	307 ms	106 ms

Profiling. Under load the proxy used about 0.86 of one core; its CPU profile was dominated by network syscalls and the goroutine scheduler, its live heap was under 4 MB, and its goroutines all mapped to known roles — the signature of a proxy that moves data rather than computing.

Chaos test. Across ten backend kills under sustained load, every client request was served, no error burst exceeded one second, and the goroutine count returned to baseline. The harness confirms that the chaos actually occurred by counting evictions from the metrics.

Goals. Table 5.6 summarises the outcome against the five goals.

Table 5.6: Results against the project goals.

Goal	Status	Evidence
G1	Met	Chaos returns to baseline; goleak green in CI
G2	Met	Reload and drain tests pass against the built binary
G3	Met	Zero error burst across ten backend kills
G4	Met	Live Grafana dashboard of backend state
G5	Met	make bench reproduces all five experiments

5.2 Discussion

The results confirm the design. The sub-millisecond overhead (E1) shows the lock-free data path is not a bottleneck; the exact linearity of E2 confirms the goroutine-per-connection model holds and that no per-connection resource leaks. E3 and E5 demonstrate that the adaptive algorithm and the health loop each deliver a large, measurable reduction in tail latency under partial failure — the situations that matter most in practice. E4 verifies the theoretical $1/N$ property of consistent hashing on the actual shipped ring. Together with the chaos result, the evidence shows the system meets all five goals not by assertion but by measurement.

An honest limitation surfaced in E5: because the active prober checks only TCP connectivity, it briefly readmits a backend that accepts connections but does not serve them, causing a small tail (p999) even under the full health loop. An HTTP-level probe would close this gap and is noted as future work.

5.3 Comparative Analysis

The comparative results follow directly from the experiments. **L4 versus L7:** L4 is roughly twice as cheap as L7 (E1), because it copies bytes while L7 parses HTTP — the expected trade-off between speed and routing intelligence. **Least-connections versus round-robin:** identical when all backends are healthy, but least-connections is five times better on mean latency when one backend is slow (E3). **Consistent hashing versus modulo:** nine times fewer keys move on a membership change (E4), the difference between a 10% and a 90% cache-miss event. **Health loop on versus off:** the difference between an 8 ms and a 106 ms mean under a sick backend (E5). In every comparison the more sophisticated choice pays for itself precisely under failure, which is when a load balancer earns its keep.

Chapter 6

Conclusions and Recommendations

6.1 Conclusions

This project designed, implemented, and evaluated LoadGate, a concurrent L4/L7 load balancer written in Go against the standard library with only four runtime dependencies. All five measurable goals were met: no goroutine leaks under sustained fault injection, no dropped in-flight requests during reload and shutdown, sub-second failover, live observability, and a fully reproducible benchmark report. The system adds under half a millisecond of latency per request, scales linearly to ten thousand connections in under 200 MB, and routes correctly around failed and slow backends.

Beyond the working system, the project demonstrates a methodology: every design decision is backed by a measurement, and the chaos-testing process uncovered three genuine correctness bugs in the failover path that ordinary unit tests missed. The lesson — that resilience must be tested by inducing failure, not assumed — is itself a result of this work.

6.2 Recommendations and Future Enhancements

The following extensions are natural next steps:

- **HTTP-level health probes** to detect backends that accept connections but do not serve, closing the gap found in E5.
- **SNI-based routing in L4 mode**, inspecting the TLS ClientHello to route by hostname while preserving end-to-end encryption.
- **HTTP/2 to backends** and WebSocket/upgrade support in L7 mode.
- **Weighted least-connections**, combining capacity weights with live load.
- **A stronger load generator** to characterise maximum throughput under saturation, complementing the latency-focused benchmarks.
- **Distributed operation** — multiple cooperating instances and shared rate limiting — for high availability beyond a single process.

Bibliography

- [1] Go standard library documentation: `net`, `net/http`, `net/http/httputil`, `crypto/tls`, `context`, `log/slog`, `hash/fnv`. <https://pkg.go.dev>.
- [2] D. Karger, E. Lehman, T. Leighton, R. Panigrahy, M. Levine, and D. Lewin, “Consistent Hashing and Random Trees: Distributed Caching Protocols for Relieving Hot Spots on the World Wide Web,” in *Proc. 29th ACM Symposium on Theory of Computing (STOC)*, 1997.
- [3] nginx development team, upstream module: smooth weighted round-robin balancing. <https://github.com/nginx/nginx>.
- [4] Envoy Proxy documentation, “Outlier Detection”; HAProxy Configuration Manual, `observe`, `rise`, `fall` directives — prior art for the asymmetric evict/readmit policy.
- [5] The Go runtime netpoller (`runtime/netpoll_epoll.go`) — basis for the goroutine-per-connection argument.
- [6] G. Tene, “How NOT to Measure Latency,” 2015 — coordinated omission and constant-rate load generation.
- [7] Prometheus and Grafana documentation. <https://prometheus.io>, <https://grafana.com>.

Appendix A

Configuration Reference

```
mode: 17                                # 14 or 17
listen: "0.0.0.0:8080"
balancer: least_connections             # round_robin | weighted_round_robin
                                         # least_connections | consistent_hash
tls:                                    # optional; presence enables HTTPS
  cert: /path/to/cert.pem
  key:  /path/to/key.pem
timeouts:
  dial: 300ms
  read: 30s
  write: 30s
  idle: 60s
  request: 30s
  drain: 15s
health:
  active: { interval: 2s, timeout: 500ms, rise: 2 }
  passive: { fall: 3, cooldown: 10s }
rate_limit:
  global_rps: 0                         # 0 = unlimited
  per_client_rps: 0
routes:                                # L7 only; longest path prefix wins
  - match: { path_prefix: "/" }
    pool: web
pools:
  web:
    - addr: "127.0.0.1:9001"
      weight: 3
    - addr: "127.0.0.1:9002"
      weight: 1
```

Appendix B

Command-Line Reference

```
loadgate check -c <file>    validate a config file and exit
loadgate run   -c <file>    serve until interrupted

run flags:
  -c                config path (default config.yaml)
  -log-level        debug | info | warn | error (default info)
  -log-format        json | text                (default json)
  -metrics-addr      metrics endpoint (default 127.0.0.1:9095)
  -debug-addr        pprof port, e.g. 127.0.0.1:6060 (default: disabled)

reload:  kill -HUP <pid>    validate-before-swap; keeps old config on error
```

Appendix C

Reproducing the Results

```
make build          # build the binary
make test           # unit tests with the race detector
make bench          # benchmarks E1-E5 -> test/bench/results/
bash test/chaos/run.sh # chaos harness (Docker)
go run test/chaos/verdict.go # grade the chaos run
bash test/bench/profile.sh # capture pprof profiles -> docs/profiles/
docker compose up --build # full demo stack (proxy, backends, Grafana)
```